

When Is a Market Estimate Trustworthy?

Error Bounds for Margin-Constrained Market Disaggregation

Abstract

Most market estimates are built by taking a published total and multiplying it down through geography, industry, and segment shares. The method assumes spend is a measure over a partitionable universe, so that any market is a subset sum. That assumption is sound. The problem is that it says nothing about whether a particular estimate is any good, and practitioners need exactly that.

We show the question has a clean answer. Iterative proportional fitting (IPF) reproduces its input margins exactly, so its residual is orthogonal to the span of the observed constraints. Only the part of a query pointing outside that span can be wrong. This gives a reliability statistic computable before you answer, with no access to ground truth. Tested against real published joint distributions from the U.S. County Business Patterns (40,560 cells, 6.1M establishments), it is calibrated to within $1.15\times$ across six seeds and cuts the dispersion of prediction error by about a third against a query-size baseline. It does *not* rank queries better than query size alone, and we report that head-to-head in full because the temptation to claim otherwise is strong and the fair comparison is easy to get wrong.

We then ask how much better modelling could do, and the answer is: not much. Three independent approaches, a parametric interaction term, low-rank completion, and covariate regression, each recover somewhere between 2% and 11% of the residual. Roughly nine tenths of it is cell-specific and out of reach of anything transferable. Retaining past estimates as constraints helps by 87% on queries near what was already observed and by an amount not distinguishable from zero on queries that are not, and the benefit saturates after a *single* nearby observation. The value of such a store therefore scales with the number of distinct neighbourhoods it covers, not with how many records it holds. Moving from establishment counts to dollars costs a steady factor of about $1.5\times$, which is a tax rather than a wall. We also document two failures that leave no trace in the output: IPF converges silently and order-dependently when its margins disagree, and disclosure control removes cells in a pattern built specifically to defeat inference from margins.

1 Introduction

A market estimate usually starts with a number someone else published and works downward. Global category total, times a country share, times a segment share, times a channel share. The logic holds up: expenditure is an accounting identity, so there is a finite pool, and any market is a piece of it. Treat spend as a measure over disjoint cells and market size becomes a subset sum, with disaggregation reduced to interpolation under a known constraint.

Two things get in the way. Markets are not cells of the census lattice, because a market is really an equivalence class under substitutability, and that cuts diagonally across geography and industry codes. More seriously, the joint distribution over cells is not published. What gets published is a handful of low-dimensional projections of it. Recovering a joint from its margins is badly underdetermined, and the usual remedy, maximum-entropy fill-in by IPF, picks one point out of a large feasible set rather than finding the right one.

So a disaggregated estimate can be excellent or worthless, and nothing about the output tells you which. That is the gap this paper closes.

What is here. We restate markets as linear functionals over spend atoms, which lets overlapping markets coexist without breaking additivity. We derive a per-query error identity and turn it into a calibrated statistic. We measure realised error against published joints rather than simulated ones. We put a ceiling on what interaction modelling can add. And we document two failure modes that current practice does not surface.

The mathematics is not new. The orthogonality in Section 3 is elementary linear algebra. IPF goes back to Deming and Stephan [1], its convergence theory to Csiszár [2], and estimation for small domains has a mature literature of its own [8]. What we offer is the packaging of an elementary fact into a decision rule, plus the calibration that makes it usable.

2 Setup

Let $\mathcal{A}$ be a finite set of *atoms*: the finest cells at which spend is defined, indexed by geography, industry, size band, and period. Atoms are disjoint and exhaustive by construction. Let $T \in \mathbb{R}_{\geq 0}^{\mathcal{A}}$ hold the true spend.

A *market* is not a cell. It is a linear functional. A query $q \in [0, 1]^{\mathcal{A}}$ gives each atom a membership weight, and the market’s size is $\langle q, T \rangle$. Fractional weights carry real content: whether convenience-store coffee belongs to the coffee-shop market is a judgement about substitution, and this is where it lives. The representation also lets two markets overlap freely, since overlap just means shared atoms rather than colliding sets. That resolves an awkward fact about published figures, which is that category TAMs across an economy sum to far more than its GDP.

An *observation* is a pair (a, y) with $\langle a, T \rangle = y$. Published marginal tables are observations whose incidence vectors are indicators of fibres. Write $V = \text{span}\{a_1, \dots, a_m\}$ for the constraint span.

Given observations, IPF returns the H minimising $D_{\text{KL}}(H \| \mathbf{1})$ subject to $\langle a_i, H \rangle = y_i$. It converges geometrically, being a contraction in the Hilbert projective metric [4], and is the same algorithm as Sinkhorn–Knopp. When the constraint set is *decomposable*, the maximum-likelihood solution has a closed form and no iteration is needed [5], which is worth checking before running a loop.

3 An exact error identity

IPF reproduces its inputs exactly: $\langle a_i, H \rangle = \langle a_i, T \rangle$ for every observed a_i . So the residual $R = H - T$ satisfies $\langle a_i, R \rangle = 0$ for all i , which is to say $R \perp V$ under the unweighted inner product. For any query q , writing P_V for orthogonal projection onto V and $q_{\perp} = q - P_V(q)$,

$$\text{err}(q) = \langle q, R \rangle = \langle q_{\perp}, R \rangle. \quad (1)$$

Only the part of a query pointing outside what you measured can be wrong. Two things follow.

Some queries are exact. If $q \in V$ then $\text{err}(q) = 0$ identically. Concretely, a query that pins some axes and integrates another away is answered exactly whenever the matching margin was observed. Asking for every construction firm in a set of counties, at any size, is free.

A statistic you can compute first. Cauchy–Schwarz gives $|\text{err}(q)| \leq \|q_\perp\| \|R\|$. The quantity that matters is the *unnormalised* norm $\|q_\perp\|$, not the alignment ratio $\|P_V q\|^2 / \|q\|^2$, which divides out precisely the scale carrying the signal. We learned this the hard way: the ratio turned out to have essentially no predictive power. Treating R as isotropic to first order, the predicted relative error is

$$\widehat{\text{err}}(q) = \kappa \cdot \frac{\|q - P_V(q)\|_2}{\langle q, H \rangle}, \quad (2)$$

with κ a single scalar calibrated once per lattice. Everything on the right is available without ground truth. The query is given, $\langle q, H \rangle$ is the estimate itself, and $P_V(q)$ comes from the constraint structure by backfitting. The statistic returns exactly zero on queries in V and shrinks as queries grow, through the denominator, so it folds two separately visible effects into one number.

4 Data and protocol

Synthetic data would let us choose the interaction strength that decides the answer, which defeats the purpose. So we used a published joint instead. The U.S. Census County Business Patterns (CBP) county file reports establishment counts by county, NAICS, and employment size class, and publishes the three-way table rather than only its margins.

CBP withholds cells for disclosure reasons. We measured 67.1% of size-class cells suppressed, and the rate climbs steadily with cell sparsity, from 9.9% for the smallest firm class to 98.5% for the largest. A withheld cell cannot serve as truth, so we carved out the largest dense, fully published sub-block: **1,014 counties** $\times$ **10 NAICS sectors** $\times$ **4 size buckets**. That is 40,560 atoms holding 6.1M establishments, 75.9% of the national total, with 0.1% zero cells and no imputation anywhere. The block is selected on non-suppression, so it over-represents the well-populated part of the economy, and every result below should be read as optimistic on that account.

We fit IPF under three margin regimes: one-way margins only, which is the independence model; the two margins actually available in practice, county $\times$ sector together with sector $\times$ size; and all three two-way margins. Then we answer randomly generated queries and compare to truth. Errors are symmetric relative error, $10^{|\log_{10}(\hat{y}/y)|} - 1$.

5 Results

5.1 Individual cells are bad, aggregates are not

Median per-atom error is 40.9% under independence, 16.2% under the realistic two-margin regime, and 12.7% with all three two-way margins. Individual cells are unreliable no matter how the margins are chosen. Aggregation is what makes the method work at all, and that shapes everything that follows.

Queries lying in V came out exact on all 90 trials, at 0.00%, which confirms Section 3 numerically as well as algebraically. For box-shaped queries, meaning the realistic market shape that constrains all three axes, error falls steeply and monotonically with the share of total mass covered (Table 1).

One result cuts against intuition. Uniformly random atom subsets were the *easiest* queries, at 0.27% median error, because independent errors cancel across many atoms. Box queries were consistently the hardest, since their selection correlates with exactly the interaction structure IPF leaves out. Real markets are boxes, so Table 1 is the curve that matters, and the scattered-query numbers are misleadingly reassuring.

Query mass share	n	Median	p90	p99
< 0.003%	262	11.39%	38.34%	95.49%
0.003–0.01%	344	6.77%	20.73%	41.34%
0.01–0.03%	412	4.39%	15.12%	25.88%
0.03–0.1%	631	2.99%	11.19%	29.88%
0.1–0.3%	651	2.34%	8.17%	16.15%
0.3–1%	729	1.52%	4.84%	11.65%
1–3%	564	0.73%	2.96%	7.23%
3–10%	353	0.39%	1.39%	3.51%
> 10%	54	0.27%	0.67%	1.53%

Table 1: Box-query error against query size, realistic margin regime. Median error swings $42\times$ across the range. Bin counts are given because the final row rests on 54 queries, so its p99 is effectively a maximum and should not be read as a quantile.

5.2 The statistic is calibrated, but does not out-rank query size

We fit κ in (2) on 300 queries and test on 300 held out. Our first comparison put it at Spearman $+0.663$ against $+0.522$ for a query-size baseline, which looked like a decisive win. It was not. That baseline had been handed an arbitrary constant instead of the same calibration the new statistic received, and was scored on a slightly different subset. Run fairly, with both predictors calibrated by the identical procedure on the identical held-out set, the picture changes (Table 2).

Predictor	All query families		Box queries only	
	Spearman ρ	sd($\log_{10}$)	Spearman ρ	sd($\log_{10}$)
$\ q_{\perp}\ /\langle q, H \rangle$	+0.721	0.551	+0.602	0.601
query size alone	+0.707	0.915	+0.600	0.810

Table 2: Six seeds, 300 train and 300 test queries each, both predictors calibrated identically and scored on the same subset. Standard deviations across seeds are 0.045 and 0.042 for ρ .

On *ranking*, the two are equivalent. The advantage of $\|q_{\perp}\|$ averages $+0.013$ across all families and $+0.002$ on box queries alone, both inside seed-to-seed noise. Anyone wanting only to sort queries by riskiness can use query size and stop reading.

The advantage that survives is in *magnitude*. Dispersion of prediction error is consistently lower, $\text{sd}(\log_{10})$ of 0.55 against 0.92 across all families and 0.60 against 0.81 on boxes, holding on every seed tested. In multiplicative terms, predictions land within roughly $3.5\times$ rather than $6.5\times$. Calibration is also steadier: the identity-based statistic stayed in $0.86\text{--}1.15\times$ across seeds while the query-size baseline wandered between $0.34\times$ and $0.82\times$.

There is also one thing query size cannot do at any calibration. It cannot return zero. Equation (2) is exactly zero for $q \in V$, correctly identifying the queries that carry no error at all, and no function of query size can express that.

For completeness: the normalised alignment ratio carries essentially no signal, which is what sent us back to the unnormalised norm in the first place.

5.3 Dollars cost a constant, not a regime change

Establishment counts are the easiest thing to disaggregate, and a method that only works on counts is not a market-sizing method. We tested two further published sources at the same county $\times$ NAICS grain: Nonemployer Statistics total receipts (2,863 counties $\times$ 12 sectors) and QCEW total annual wages (668 counties $\times$ 12 sectors), holding lattice, protocol, and query distribution fixed so only the measured quantity changes.

The penalty is a steady multiplicative factor near $1.5\times$ in every query-size bin: $1.34\text{--}1.63\times$ for receipts and $1.23\text{--}2.22\times$ for wages. Nothing pathological happens. The mechanism shows up in dispersion, where the within-county sector-share coefficient of variation rises from 0.39 for counts to 0.58 for receipts, and from 0.43 to 0.73 for wages. Since error falls sharply with query size, a $1.5\times$ penalty is bought back with roughly $3\times$ more span.

5.4 How much modelling can add

Is the residual modellable rather than merely bounded? Its spectrum says there is something there: the leading component of the log-residual explains 24.3% of variance, against 3.6% for a matched Gaussian benchmark. But it does not generalise. Table 3 collects three attempts.

Method	Metric	Value
Parametric interaction, transferred to held-out counties	median-error reduction	9.4%
Same form fitted in-sample (bound for that form)	median-error reduction	10.7%
Low-rank completion (best at rank 1)	out-of-sample R^2	0.118
County covariates (log size, sector diversity)	out-of-sample R^2	0.041

Table 3: Three approaches, two metrics. The transferred figure is a mean over five train/test splits (sd 1.2, range 8.2–11.5%). The R^2 figures describe explained variance of the log-residual and are not directly comparable to a median-error reduction; converted to dispersion reduction they correspond to about 6.1% and 2.1%. All four therefore land in a single-digit band, 2% to 11%.

We flag the metric distinction because it is easy to get wrong. Explained variance and median-error reduction are different quantities, and reporting one beside the other invites a comparison that is not valid. Once converted onto a common footing, the four results agree closely, which is a good deal more convincing than any single number would have been: roughly nine tenths of the residual is cell-specific.

The transfer itself works well. Of whatever the parametric form can capture, around 82% survives being carried to held-out counties, so dimensionless structure does move across geographies. There is simply not much of it to move. Low-rank completion degrades to negative R^2 by rank 5, the usual signature of fitting noise particular to the observed cells.

5.5 Accumulated observations: local, and saturating

If each past estimate is retained as a constraint, does the system improve? We stored box observations and evaluated on held-out queries, first confirming that stored constraints were satisfied to 0.000% residual so a null result could not be blamed on a bad fit.

Proximity decides everything. With 200 observations stored, error on unrelated random markets moved from 2.57% to 2.46%; on markets neighbouring something already stored it fell from 1.49% to 0.16%, with p90 dropping from 9.26% to 3.69%. Across five seeds the related gain is $87.2\% \pm 0.9$,

about the most stable figure in this paper, while the unrelated gain is $4.8\% \pm 3.2$ and went negative on one seed. The second number is not distinguishable from zero.

Two further curves determine what a store is actually worth. Table 4 varies how many observations sit near the query.

Observations stored nearby	0	1	2	3	5	8	13	21
Median error	2.26%	0.24%	0.17%	0.13%	0.15%	0.30%	0.17%	0.16%
Gain over none	n/a	89%	93%	94%	93%	87%	93%	93%

Table 4: The first nearby observation captures essentially all of the available gain. The second and third add a sliver. Beyond that the series is flat and its wobble is noise.

Table 5 instead fixes eight stored observations and moves the target away from them.

Atom overlap with stored region	100%	95%	88%	75%	50%	25%	2%
Median error	0.09%	0.24%	0.62%	1.09%	1.06%	1.50%	1.54%
Gain over none	97%	90%	74%	61%	55%	6%	none

Table 5: Benefit survives to roughly 50% overlap and collapses between 50% and 25%.

Together these say something sharper than locality alone. Because around 51% of a typical query already lies inside the margin span, a stored observation can only speak to the remainder, and it does so within a neighbourhood of roughly 50% overlap. Within that neighbourhood one observation is nearly sufficient.

The value of such a store therefore scales with the number of distinct neighbourhoods covered, not with the number of records held. A thousand observations of one market are worth about what three are worth. Three hundred observations spread over three hundred neighbourhoods are worth roughly a hundred times more than three hundred concentrated in one. This inverts the intuition that a concentrated query stream is the favourable case: concentration saturates, and diversity is the efficient direction per record. It also means record count is the wrong health metric for such a system, and coverage is the right one.

6 Two failures that leave no trace

6.1 Margins that disagree

IPF assumes its margins are true and mutually consistent. Margins pulled from different sources, vintages, and definitions are neither. When they conflict, IPF does not oscillate and does not diverge. It settles on a fixed point of the composite map that satisfies whichever margin was applied *last* and approximates the other. Sweep-to-sweep drift reaches 10^{-16} , which is indistinguishable from clean convergence.

The practical effect is that loop order becomes an undocumented ranking of how much you trust each source. With one margin at 2% noise and another at 50%, the two orderings produced answers differing by 22.6% of total mass, and both reported convergence. Weighting each constraint by its precision removes the order dependence and cuts median error from 41.7% to 28.1%, while costing nothing when sources are equally good.

Margin noise on its own turns out to be second-order: 10% multiplicative noise moved median cell error only from 16.2% to 21.3%. The damage comes from treating unequal sources as equal, not

from noise as such. Where noisy estimates exist at several aggregation levels at once, IPF is the wrong tool entirely and optimal reconciliation is the right one, since it projects incoherent forecasts onto the coherent subspace using the full error covariance and handles non-nested groupings [9].

6.2 Suppression is adversarial by design

The 67% suppression we ran into is not incidental thinness in the data. Agencies apply *primary* suppression to sensitive cells, then *complementary* suppression to further cells chosen specifically so the primary values cannot be reconstructed from the published margins [10]. Inference from margins is the stated threat model. This is an information barrier rather than a modelling problem, and it falls hardest on small, specific queries, which Table 1 already identifies as the least reliable regime. The two effects compound.

7 Discussion

A rule you can operate. Section 5.2 means the operating rule can be stated on query size alone, which is the simpler instrument; the identity buys a tighter magnitude and the ability to certify exact queries, not a better ordering. Above roughly 1% of the anchor’s mass, median error runs below 1% and p90 below 3%. Below roughly 0.01%, p90 passes 15% and p99 approaches 100%. Since (2) can be evaluated before answering, the honest response in the second regime is to decline rather than to print a number with decimals on it. For dollar-denominated queries the thresholds move by about $3\times$ and κ needs its own calibration.

Where the leverage actually is. Ranking every intervention we measured: observing an *uncovered* cell cuts error by 87%; combining sources by precision, through a Fay–Herriot shrinkage estimator, by 33.5%; better structural modelling by 2% to 11%. The qualifier matters, because Table 4 shows a second observation of an already-covered cell buys almost nothing. Effort belongs in acquisition and in fusing evidence, not in a smarter interpolator. One implementation note, learned by getting it wrong: because market quantities span orders of magnitude, shrinkage has to run on a scale where variances are roughly homoskedastic. Fitted on the raw scale it was *worse* than either of its inputs at high noise. In log space it beat both at every noise level we tested.

Coherence is not accuracy. Reconciliation forces internal consistency whether or not the result is true. Any system that catches defects by spotting contradictions loses that signal the moment a reconciliation layer is added, and has to replace it with accuracy checks. The held-out marginal protocol used here is one such check, and it can be run continuously rather than once.

What this does not cover. Our ground truth is a dense sub-block selected on non-suppression, which biases everything optimistic. The lattice is coarse at 10 sectors and 4 size bands, far coarser than a real market definition needs. Receipts and wages are proxies for spend, and neither separates price from frequency. Establishment populations are not the buyer population for most products, and the lattice indexes *sellers*. For a local business the seller’s geography and the buyer’s coincide, which conceals the distinction; for a business selling online they are unrelated, and a pipeline that silently uses the seller’s county as the market’s geography will produce a well-formed answer to the wrong question. Our queries are boxes with binary membership, whereas real market definitions carry fractional weights that are themselves estimated and probably contribute more error than

anything we measured. And none of this touches market definition: whether two products compete is not a statistical question, and no amount of lattice resolution will answer it.

8 Conclusion

Disaggregating a published total is a sound procedure whose reliability varies by more than two orders of magnitude across the queries people actually ask, with nothing in the output to distinguish the good cases from the bad. The residual of a margin-constrained fit is orthogonal to what was measured, which makes the reliability of any particular query computable in advance. That statistic will not rank queries better than their size does, but it is better calibrated, it identifies exactly answerable queries, and it can be evaluated before committing to an answer.

The negative results are the more useful half. Modelling the residual is capped near a tenth of it. Accumulating observations helps only nearby and saturates after one. Suppression is engineered against precisely this kind of inference. Taken together they point away from better estimators and toward acquiring observations in uncovered neighbourhoods and fusing sources by measured precision, which is roughly what the federal statistical system already does.

9 Related work

IPF is due to Deming and Stephan [1], with convergence to the I -projection established by Csiszár [2] and the definitive treatment in Rüschendorf [3]. The algorithm coincides with Sinkhorn–Knopp, and the optimal-transport literature supplies sharp geometric convergence rates [4] along with the log-domain stabilisation needed when margins approach zero. Decomposable constraint sets admit closed-form solutions [5].

Applied work on spatial microsimulation reports the same sensitivity to sparsity we observe, and notes that a quality measure for the resulting estimates is generally unavailable [6]. That literature validates against aggregate constraint totals rather than a held-out joint, which is the one place our protocol improves on it. Estimation for domains too small to measure directly is the province of small-area estimation, from Fay and Herriot [7] through the survey of Rao and Molina [8]; it supplies principled mean-squared-error estimates and spatial borrowing of strength, and subsumes much of what we build here. Reconciling incoherent hierarchical forecasts is treated by Wickramasuriya et al. [9]. When constraints are both noisy and insufficient, generalised maximum entropy is the natural setting [11].

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